A Unified Recurrent Circuit Model of Normalization and Sensory Adaptation

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Abstract

Abstract

1 Introduction

To respond efficiently to stimuli, primary visual cortex (V1) dynamically regulates overall activity and adjusts population encodings based on recent stimulus history. The normalization model was developed to explain part of this regulatory behavior and posits that cortical neurons use recurrent feedback loops to dynamically adjust the overall response magnitude [1, 2]. Adaptation is the complimentary phenomenon that alters neural encodings according to recent history and has been shown to function as an efficient coding mechanism [3–5]. Unlike normalization, the effects of adaptation are much less well-defined. Many studies present seemingly contradictory results that strongly depend on experimental conditions.

Despite the ubiquity of normalization in the visual cortex and the many existing models of sensory adaptation [3, 5–11], little effort has been made to reconcile the two into a joint theoretical framework. The main exception to this is presented in [8], where the authors develop a model of adaptation that dynamically adjusts the normalization weights between pairs of neurons. However, modifying synaptic weights is typically a relatively slow process, making it a less plausible mechanism for cortical adaptation that occurs on the order of hundreds of milliseconds and is rapidly reversible [12]. Furthermore, continuous weight remapping risks network instability whereas single-neuron gain modulation offers a fast and reversible mechanism that operates on a time-scale better suited for rapid adaptation [13]. Importantly, the normalization reweighting model [8] was unable to explain contrast-dependent changes due to adaptation, such as changes in contrast gain.

We propose a unified dynamical model of concurrent normalization and adaptation that relies strictly on the modulation of neural gains. Our framework combines ORGaNICs, a normalization model in which activity-pooling interneurons dynamically regulate recurrent feedback [14–16], with an adaptation mechanism that maintains covariance homeostasis [6, 7]. The model couples fast normalization that pools over a spatially broad region of the visual field with a slower adaptive gain modulation that is restricted to local regions of the visual field. Within the adaptation circuit, interneurons measure principal cell activity along multiple axes in population space, adjust their gains according to online estimates of marginal variances along these directions, and provide subtractive feedback to factorize a homeostatic covariance transformation. At its fixed point, the circuit captures the complex interplay between normalization and adaptation while preserving the core functions of both mechanisms.

We show that our model reproduces a variety of experimental observations, including response gain modulation, contrast gain shifts, and shifts in tuning. For simplicity and to compare directly with leading experimental literature, we model adaptation to oriented visual gratings throughout our analysis. We first adapt our circuit to a single, prolonged adaptor stimulus to investigate the interplay between adaptation and normalization. When the adaptor is small, adaptation drives tuning suppression and repulsion that intensify over time. When the adaptor is large, we observe transient tuning suppression, followed by attraction and negligible change in suppression. Next, we adapt our network to ensembles of orientation inputs under biased statistics, showing that tuning curves of adapted neurons follow a similar pattern of suppression found in experiment [17]: weak untuned suppression and strong suppression at the orientation of the adaptor. We demonstrate that our model obeys an empirically observed power law that emerges directly from efficient coding principles [5, 18]. Finally, we test this power law outside of experimentally verified regimes and show that our model predicts a normalization-induced breakdown of this power law behavior. Our framework provides a mathematically motivated, mechanistic understanding of how normalization and adaptation interact to dynamically shape population encodings across diverse sensory contexts.

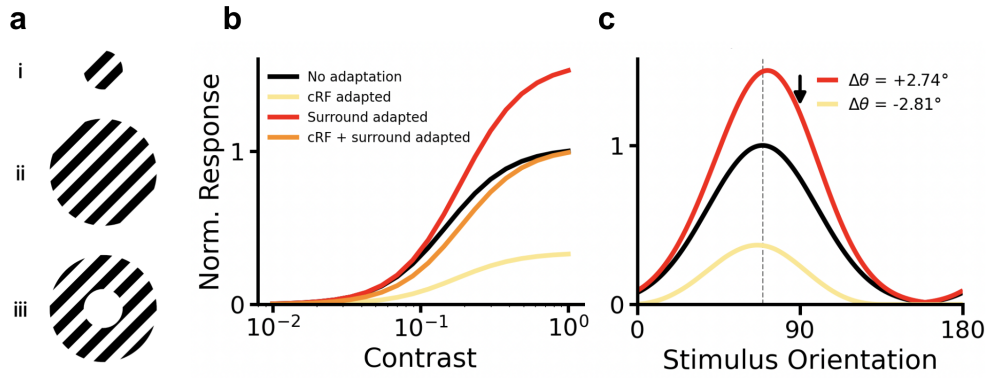


Fig. 1 Interplay of adaptation and normalization. a: Three oriented grating configurations used during adaptation, spanning (i) the center of the visual field, (ii) both center and surround, and (iii) the surround only. **b:** Contrast response functions for a population of neurons with central RFs, evaluated for test gratings presented at the adaptor orientation. **c:** Tuning curves for a neuron with an orientation preference adjacent to the adaptor. Black arrow, adaptor orientation. Dotted vertical line, preferred orientation in the unadapted state. For both **b** and **c**, test gratings were large regardless of the spatial extent of the adaptor. Black curves, unadapted baseline. Yellow curves, adaptation to center-only stimuli (i). Orange curves, adaptation to stimuli covering both center and surround (ii). Red curves, adaptation to surround-only stimuli (iii).

2 Results

2.1 Adaptation to center and surround

Experimental evidence suggests that cortical normalization operates across a broad region of space (e.g., about 4x the receptive field diameter) by amplifying weak inputs more than strong inputs via recurrent excitation [19, 20]. Adaptation, on the other hand, operates by suppressing activity locally (e.g., within a region roughly equal to an adapted neuron's classical receptive field) [21, 22]. This gap in the spatial extent of normalization and adaptation allows us to probe their interplay by varying the sizes of stimuli presented during adaptation.

We used our circuit model to simulate adaptation to oriented gratings with three different spatial spans (Figure 1a): small, large, and surround adaptors, centered on a simulated neuron's receptive field (RF). The small adaptor evoked activity only in the neuron's classical receptive field (cRF) whereas the surround adaptor avoided the cRF an the large adaptor covered both the cRF and the surround. In each case, we presented the circuit with a prolonged, vertical (or near-vertical) adaptor stimulus at high contrast ($c = 80\%$). Using gain values taken from the end of the ($t = 2.1$ sec) adaptation period, we calculated steady-state responses to test stimuli analytically using Equations 10 and 11 (see Methods). The test stimuli always covered both the cRF and surround regardless of the size of the adaptor. As a note on terminology, a classical receptive field (cRF) and its surround denote functional driving and modulatory zones for a neuron, regardless of absolute visual field coordinates.

Adaptation changed either the contrast gain or the response gain of a neuron's activity, depending on the size of the adaptor (Figure 1b), consistent with experimental results [23–25]. We simulated adaptation to small and large stimuli and compared the resulting responses to a non-adapted baseline (black curves in Figure 1). The adaptor and test stimuli were centered on a neuron's RF and matched the neuron's orientation preference. For each adaptation case, we calculated the response of the neuron to large test stimuli over a range of contrasts between 0 and 1. When adaptors were small (restricted to the neuron's cRF), local adaptive suppression resulted in an overall decrease in response gain (Figure 1b, yellow curve). Conversely, when the adaptor was large and included the surround, the combination of local adaptation and surround suppression from normalization caused a rightward shift of the contrast response function and a decrease in contrast gain (Figure 1b, orange curve). Within the context of the normalization model, this rightward shift corresponds to an increase in the effective semi-saturation constant σ in Equation 2.

Adaptation of the surround resulted in "facilitation" (Figure 1b, red curve), consistent with experimental evidence [25]. We simulated surround-only adaptation by driving neurons with RFs in the surround with an oriented stimulus (Figure 1a, iii). Adaptation resulted in reduced activity among these surround-responsive neurons. When we tested our model using large, high-contrast test stimuli, neurons's with RFs centered on the test stimulus received weaker normalization signals from the surround. Consequently, responses to the test stimuli were facilitated above the non-adapted baseline due to this release from surround suppression.

Adaptation shifted orientation tuning curves, resulting in either "tuning repulsion" or "tuning attraction" depending on the adaptor (Figure 1c), consistent with experimental results [26]. To test this, we varied the spatial extent of a high contrast adaptor stimulus and computed responses to large test stimuli with the same high contrast as the adaptor but with many orientations. Then, we calculated the peak of the orientation tuning curve and the shift in tuning ($|\theta_{adapt} - \theta_{pref}| \approx 15°$) before and after adaptation. For adaptation induced by a small stimulus (Figure 1c, yellow curve), simulated neural responses exhibited tuning repulsion, in which their peak orientation preferences shifted away from the adaptor. This occurred because adaptation suppressed responses particularly for stimuli near the adaptor orientation, leaving less-suppressed orientations to dominate responses to test stimuli. In contrast, adapting only the surround (Figure 1c, red curve) produced tuning attraction, where the peak orientation preference shifted toward the adaptor. This occurred because facilitation was strongest for test stimuli close to the adaptor orientation, where surround activity was most heavily suppressed. Whether adaptation to large stimuli (Figure 1a(ii)) resulted in tuning attraction or repulsion depended on the duration of adaptation. In the next section, we investigate whether our model is consistent with experimental findings regarding this temporal contingency.

2.2 Temporal dynamics of adaptation

To further investigate the coupling dynamics of adaptation and normalization in our model, we simulated an adaptation experiment that systematically varied adaptor size and duration [27]. We added a short delay ($t_{delay} = 40$ ms) in normalization signals

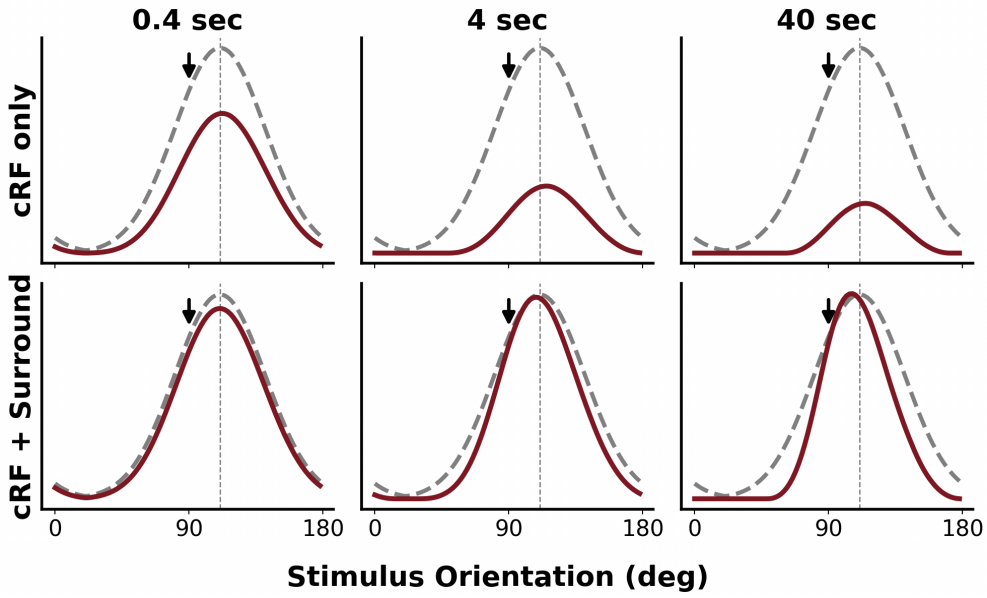


Fig. 2 Effects of adaptation depend on adaptor size and duration. Each panel displays the original tuning curve (dashed) and modulated tuning curve (red) of a neuron that prefers an orientation adjacent to the adaptor (vertical dotted line). The top row corresponds to tuning curves that have been adapted to stimuli that only cover the cRF, and the bottom row corresponds to adaptation to stimuli that cover both the cRF and surround. Columns indicate the duration of adaptation, increasing from left to right. This plot is inspired by Figure 1 in Ref. [27].

from the surround consistent with experimental evidence for such a delay [28–30]. We followed the same adaptation protocol as in the previous section by presenting one orientation at high contrast ($c = 80\%$) for a sustained period. We then extracted gain values at three different times during adaptation ($t = 0.4, 4, 40$ sec) and used them to calculate steady-state responses to test stimuli via Equations 10 and 11. For each adaptor (cRF or cRF + surround), we computed the responses of a neuron to test stimuli with full contrast ($c = 100\%$) and many orientations to visualize the tuning curves in Figure 2.

When the adaptor orientation was on the flank of a neuron's preferred orientation, adaptation to small stimuli induced tuning suppression and repulsion. These tuning shifts intensified with adaptation duration (Figure 2, top row), consistent with experimental evidence [27]. In our circuit, the small adaptor drove high-variance responses that triggered suppressive interneuron feedback. By Equation 8, the rate of change of interneuron gains is proportional to online estimates of the variance of the responses, so gain changes stagnated as marginal variances reach their target values. Indeed, we observed that the suppression and repulsion intensified quickly at first and then slowly approached steady state, i.e., the difference in responses for short and medium duration adaptors in the top row of Fig. 2 is greater than the difference for medium and long duration adaptors.

Adaptation to large stimuli produced tuning suppression with negligible shift in tuning for brief adaptors, but tuning attraction and negligible suppression for prolonged adaptors, consistent with experimental findings [27]. The large adaptor drove high response variance across many neurons, causing our circuit's gain control mechanism to adapt accordingly. However, for short duration adaptors, neurons responding to the test stimuli received non-adapted normalization signals from the surround because of the delay in normalization signals from the surround. For a brief moment, each principal cell behaved as if it was adapting to a small stimulus instead of a large one, resulting in slight suppression for very short adaptation periods (Figure 2, bottom left). Although slight repulsion occureds during the very short adaptation period, it was no longer noticeable at $t = 0.4$ seconds (Figure 2, bottom left). However, following long adaptation durations, responses to test stimuli reflected adapted normalization signals from the surround. These weaker normalization signals resulted in greater facilitation of the responses to test stimuli with orientations near the adaptor, yielding in an attractive shift of the tuning curve toward the adaptor over time (Figure 2, bottom middle and right). Additionally, since the test stimulus was high contrast, the responses saturated at a very similar level with and without adaptation (Figure 2, bottom right).

Neither adaptation nor normalization alone is sufficient to account for the wealth of effects outlined above. A joint understanding of the two phenomena is especially required to understand how adaptation to large stimuli can affect subsequent responses. Our framework is the first of its kind to capture all of these phenomena through principled dynamical equations. However, these results are not unique to a covariance-homeostasis adaptation mechanism. In fact, any properly coupled model that features local suppression to an adaptor and broader normalization (with a delay) would be sufficient. The following sections therefore focus on the implications and performance of our specific adaptation mechanism combined with normalization.

2.3 Homeostasis via adaptation [FLAG]

In order to better understand the adaptive behavior of our model, we simulated adaptation to inputs that are drawn from ensembles of oriented stimuli rather than one prolonged stimulus. First, we simulated the seminal experiments of Benucci et al. [17] in which they presented sequences of large orientation gratings drawn from a biased or uniform ensemble in rapid succession (32 ms each) to anesthetized cats [17]. In the uniform ensemble, all orientations were equally likely. In the biased ensemble, one orientation was presented with a much higher probability than the rest (Figure 3a,b). The authors deviated from previous studies by measuring responses *during* adaptation rather than by probing the network with test stimuli afterward [17]. Since surround interactions in V1 have been shown to exhibit a delay [28–30], the extremely rapid nature of orientation changes likely decoupled surround and cRF signals from one another. In other words, the input drive to a neuron and the normalization signals it received from the surround corresponded to different stimuli in the rapid sequence. For this reason, the authors observed effects that are otherwise consistent with adaptation to small stimuli [17, 27].

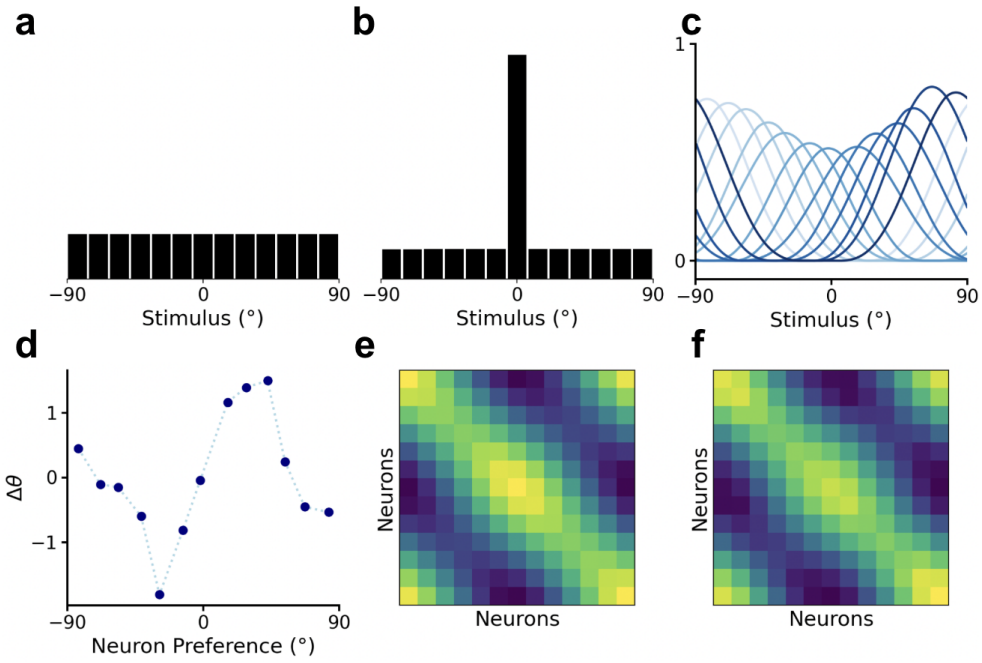


Fig. 3 Adaptation to stimulus ensembles. a: Probability histogram of oriented inputs for the uniform ensemble. **b:** Probability histogram of oriented inputs for the biased ensemble where the adaptor orientation (0°) is shown 1/3 of the time. **c:** Tuning curves of simulated V1 neurons after adaptation. **d:** Tuning shifts after adaptation with respect to original neuron orientation preference. **e:** Covariance matrix of the biased ensemble of input drive to the V1 neurons. **f:** Covariance matrix of adapted responses to the biased input ensemble.

For consistency with the experiment, we measured responses during exposure to sequences of stimuli (contrast $c = 80\%$), drawn from a uniform or biased set of 26 evenly spaced oriented stimuli. We recorded responses to each stimulus at the last time-step of exposure and took the average across multiple presentations of the same stimulus to calculate tuning curves. The tuning curves were normalized by setting the maximum and minimum response in the uniform ensemble to one and zero respectively.

Simulations of our model reproduced experimental observations of a slight reduction in responses for orientations orthogonal to the adaptor (i.e., the biased orientation) and a much more pronounced suppression effect near the adaptor [17]. These effects are visible in Figure 3c. In our circuit, biased stimulus statistics drive high response variance in neurons tuned to the adaptor orientation. Adaptive gain control scales interneuron feedback to reduce this variance, resulting in strong suppression near the adaptor orientation. Less intuitively, high variance also arises in neurons tuned to orthogonal orientations (see the corners of Figure 3e). Because these neurons remain silent during frequent adaptor presentations but fire strongly during rare orthogonal stimuli, their intermittent activity results in high variance that triggers adaptive suppression.

During adaptation, responses exhibited a tendency toward equalization in both first- and second-order statistics, matching experimental observations [17]. Adaptation drives the network towards first-order equalization by providing heavier suppression to adaptor-tuned neurons than flank and orthogonal neurons, flattening the average population response profile. Second-order equalization occurs as adaptive feedback selectively dampens high-variance response directions, reducing co-activity among strongly driven neurons and decorrelating output responses relative to the input covariance structure (seen as the decreased power in high-variance regions from Figure 3e to Figure 3f). These homeostatic effects are robust across time, model parameters, and stimulus parameters, varying only in magnitude. The exception is that the delay of surround normalization signals must be greater than the rate of stimulus presentations ($t_{delay} > 32$ ms).

2.4 Adaptation magnitude follows a power law

We extended our analysis of ensemble adaptation by comparing simulated responses with recent experimental findings in mouse V1 [18], which demonstrate that adapted population responses obey a power law relative to stimulus probability. Specifically, the ratio of adapted response magnitudes across two environments scales as a power law of the ratio of stimulus probabilities [5, 18]:

$$\frac{\boldsymbol{r}_X(s_i)}{\boldsymbol{r}_Y(s_i)} = \left[\frac{\boldsymbol{p}_X(s_i)}{\boldsymbol{p}_Y(s_i)}\right]^{\beta} \tag{1}$$

where X and Y denote two environments with different stimulus statistics, $\boldsymbol{r}_X(s_i)$ and $\boldsymbol{r}_Y(s_i)$ are the responses to stimuli s_i in each environment, and $\boldsymbol{p}_X(s_i)$ and $\boldsymbol{p}_Y(s_i)$ are the stimulus probabilities. In plain terms, this means that relative shifts in stimulus frequency across contexts map onto proportional, predictable adjustments in neural activity.

The authors found very consistent power law behavior in the range $\beta \in [-0.4, -.15]$ across trials that tested probability ratios within $log_{10}\left[\boldsymbol{p}_X(s_i)/\boldsymbol{p}_Y(s_i)\right] \in [-1, 1]$ [18]. Intuitively, the exponent β is negative because adaptation causes more probable stimuli regions to evoke weaker responses.

The very same power law emerges from our model when for the same range of stimulus probability ratios. We adapted our network by randomly drawing stimuli from three different Von Mises distributions (Figure 4, left). During adaptation ($t = 4.2$ sec), the network was presented with small, high contrast ($c = 80\%$) stimuli drawn from three ensembles (Figure 4) and presented for 30 ms each. Within the range of stimulus probability ratios $log_{10}\left[\boldsymbol{p}_X(s_i)/\boldsymbol{p}_Y(s_i)\right] \in [-1, 1]$, we found that responses of our model produce a strong fit ($R^2 = 0.91$) to the power law with $\beta = -0.31$ (See the dotted line in Figure 4). This exponent falls almost directly in the center of the experimentally determined region [18]. This was not a designed feature of the model, but rather an emergent phenomenon.

Consistency with this power law indicates that our mechanistic model aligns with normative principles of efficient coding. Indeed, this relationship has been shown to reflect an optimal balance between representational fidelity and metabolic cost-efficiency [5]. By coupling input decorrelation via adaptation with dynamic range

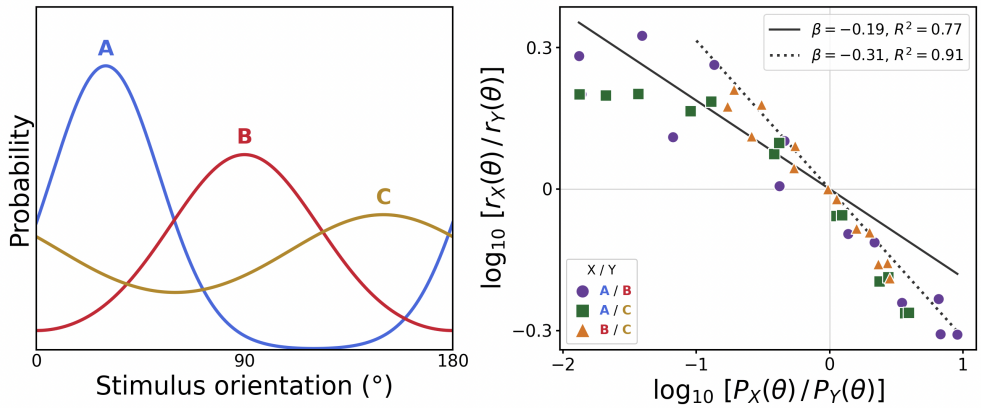


Fig. 4 Adaptation magnitude across stimulus statistics. Left: Adaptation-phase Von Mises stimulus probability distributions. **Right:** Ratio of response magnitudes as a function of the ratio of stimulus probabilities across statistical contexts. Dashed line is a power law fit in the logarithmic domain $[-1, 1]$. Solid line is a power law fit for the full domain $[-2, 1]$.

control via normalization, our model uses biologically grounded dynamics to reproduce efficient coding principles previously captured through ad-hoc models.

Furthermore, simulations outside of the measurement range of the original experiment predict a testable breakdown of this power law at more drastic probability ratios. Upon expanding the range of log probabilities from $[-1, 1]$ to $[-2, 1]$, R^2 drops from 0.91 to 0.77, indicating a failure to obey the power law (see the black curve in Figure 4). This arises because normalization puts an upper bound on response magnitude, even to very infrequent stimuli. So, even when stimulus probabilities are drastically different, normalization does not allow their magnitudes to deviate to an equally extreme extent. This breakdown of the power law is intuitively more pronounced for stimuli that activate more normalization signals, such as large stimuli.

3 Methods

3.1 Overview

We developed a dynamical model of concurrent normalization and adaptation to model information processing in primary visual cortex (V1). To do so, we combined ORGaNICs, a model of normalization developed by Heeger et al. [15, 16] with a model of adaptation inspired by Duong et al [6]. We designed the model so that it achieves a fixed point that is consistent not only with both frameworks individually, but also with a dynamic coupling that maintains a normalized state throughout adaptation.

ORGaNICs features a pool of interneurons that modulate recurrent feedback gain to achieve steady-state responses that obey the normalization equation [15, 16]:

$$\text{Normalized Response} = \frac{\boldsymbol{z}^n}{\sigma^n + \sum_i \boldsymbol{z}_i^n} \tag{2}$$

where $\boldsymbol{z} \in \mathbb{R}^N$ is the input and σ is the "semi-saturation constant", defined as the contrast for which responses reach half of their maximum value. Generally, models of the visual cortex use $n = 2$ as the exponent of normalization to match experimental observations [31].

Adaptation in V1 is hypothesized to optimize representation fidelity and metabolic cost [3–5]. The adaptation model formulated by Duong et al. [6] does so by relying on interneuron feedback to achieve its fixed point. Specifically, it deploys an overcomplete set ($K \geq \frac{N(N+1)}{2}$) of interneurons that estimate many marginal variances [6, 7]. Interneuron feedback is multiplied by gains that act as Lagrange multipliers to factorize a covariance transformation on the inputs:

$$\text{Adapted Response} = \left[I + W_{yv} D(\boldsymbol{g}) W_{yv}^\top\right]^{-1} \boldsymbol{z} \tag{3}$$

where $W_{yv} \in \mathbb{R}^{N \times K}$ is the synaptic weight matrix that connects principal neurons to interneurons, and $D(\boldsymbol{g}) \in \mathbb{R}^{K \times K}$ is a diagonal matrix with the interneuron gain vector $\boldsymbol{g} \in \mathbb{R}^K$ along the diagonal. The gains adapt according to second-order stimulus statistics so that the matrix in brackets in Eqn. 3 transforms the input covariance towards a homeostatic target.

A schematic of the joint model is shown in Figure 5. Simulated inputs carrying contrast and orientation information from the LGN provide drive to the V1 model and are dynamically adapted and normalized to form the response. Spatially, we model V1 neurons with receptive fields centered across multiple locations (represented by black circles at the bottom of the figure), where each subpopulation of neurons with overlapping receptive fields undergoes local adaptation and all neurons contribute to a global normalization pool (see "Modeling the Surround").

3.2 Full Equations

In this section, we present our dynamical firing rate model for joint normalization and sensory adaptation in V1. Neural activity is formatted as vectors, where each component corresponds to an individual neuron's response (either membrane potential or firing rate). In our notation, boldface lowercase letters indicate vectors, boldface capital letters indicate synaptic weight matrices, and all other terms indicate constants. The model includes a set of principal neurons ($\boldsymbol{y}$), three sets of interneurons ($\boldsymbol{u}$, $\boldsymbol{a}$, $\boldsymbol{v}$), and a set of interneuron gains ($\boldsymbol{g}$).

The response of the principal neurons is a sum of input drive, recurrent feedback weighted by normalization pool neurons, and feedback from gain-modulated interneurons (also weighted by the normalization pool). The full dynamical equations are as follows:

$$\tau_y \frac{d\boldsymbol{y}}{dt} = -\boldsymbol{y} + \beta \cdot \boldsymbol{z} + \left(\frac{1}{1 + \boldsymbol{a}^+}\right) \boldsymbol{W}_{yy} \sqrt{\boldsymbol{y}^+} - \left(\frac{\boldsymbol{a}^+}{1 + \boldsymbol{a}^+}\right) \boldsymbol{W}_{yv} \boldsymbol{v} \tag{4}$$

$$\tau_u \frac{d\boldsymbol{u}}{dt} = -\boldsymbol{u} + \left(\frac{\sigma}{2}\right)^2 + \boldsymbol{N}(\boldsymbol{y}^+ \odot \boldsymbol{u}^{+^2}) \tag{5}$$

$$\tau_a \frac{d\boldsymbol{a}}{dt} = -\boldsymbol{a} + \boldsymbol{u}^+ + \boldsymbol{a} \odot \boldsymbol{u}^+ \tag{6}$$

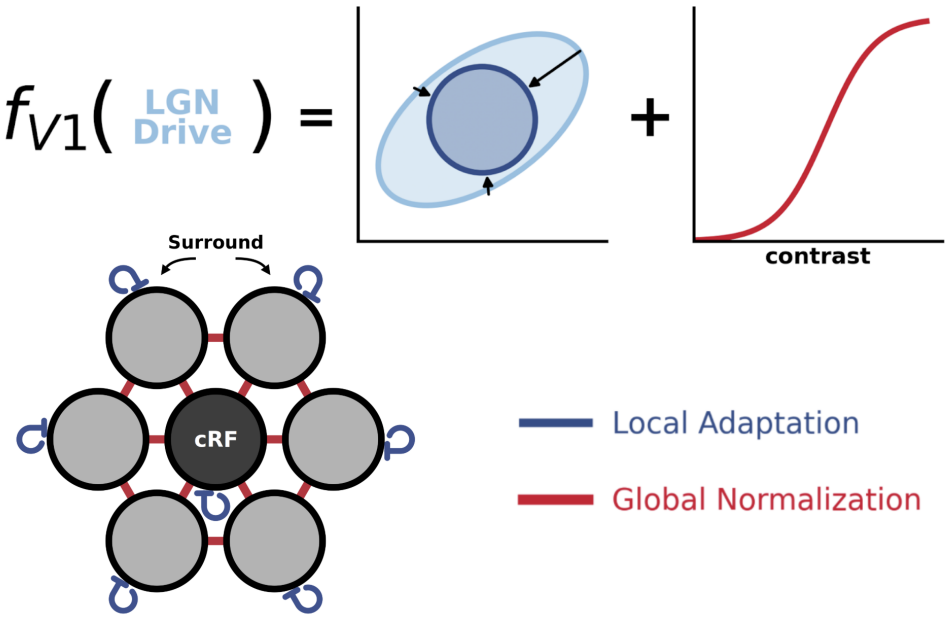


Fig. 5 Model Schematic. LGN drive to V1 is concurrently adapted and normalized by our model. The ellipse deformation into a circle represents adaptation via a covariance transformation. The red sigmoid represents the non-linear impact of normalization on responses to contrast-scaled inputs. The schematic on the bottom represents the center-surround spatial structure of the model. Each black circle is one receptive field (RF). Blue circles connecting RFs to themselves represent local adaptation, and the connecting web of red lines indicate global normalization.

$$\tau_v \frac{d\boldsymbol{v}}{dt} = -\boldsymbol{v} + \boldsymbol{g} \odot \boldsymbol{c} \tag{7}$$

$$\tau_g \frac{d\boldsymbol{g}}{dt} = (\boldsymbol{c} - \boldsymbol{\mu})^2 - \boldsymbol{\theta_t} \tag{8}$$

$$\tau_\mu \frac{d\boldsymbol{\mu}}{dt} = -\boldsymbol{\mu} + \boldsymbol{c} \tag{9}$$

where the membrane potentials of the principal cortical neurons are denoted by the vector $\boldsymbol{y}$, and their corresponding firing rates are represented by $\boldsymbol{y}^+$. The input current to the adaptation interneurons $\boldsymbol{v}$ are denoted as $\boldsymbol{c} = \boldsymbol{W}_{yv}^\top \sqrt{\boldsymbol{y}^+}$, computed as a weighted sum of the principal firing rates. The "+" superscipt indicates rectification and is used to estimate firing rates from membrane potentials. Specifically, firing rates are estimated by rectifying and then raising to an exponent ($\boldsymbol{x}^+ = \lfloor \boldsymbol{x} \rfloor^\alpha$, where $\alpha = \{2, \frac{1}{2}, 1\}$ for $\boldsymbol{x} = \{\boldsymbol{y}, \boldsymbol{u}, \boldsymbol{a}\}$) [15, 16, 31]. The variable $\boldsymbol{z}$ represents the input drive to V1 computed as weighted sums of the firing rates of lateral geniculate nucleus (LGN) neurons. The input gain denoted by β is set to a constant ($\beta = \frac{1}{2}$) for the purposes of this paper, although input gain more generally is hypothesized to model the effects of attentional modulation [15, 16]. The matrix $\boldsymbol{W}_{yy} \in \mathbb{R}^{N \times N}$ in the third term

in Equation 4 represents the recurrent synaptic weight matrix that connects principal cells to themselves. The Hadamard product "$\odot$" signifies the element-wise multiplication of vectors. Time constants are denoted by τ, with subscripts corresponding to each dynamic variable (refer to the table below for values).

τ_y	τ_u	τ_a	τ_v	τ_g	τ_μ
0.2 ms	15.0 ms	0.1 ms	15.0 ms	2.0 s	2.0 s

The dynamic variables $\boldsymbol{u}, \boldsymbol{a} \in \mathbb{R}^N$ are inhereted from the ORGaNICs framework [15, 16] and represent the membrane potentials of interneurons that compute the normalization signals. The matrix $\boldsymbol{N} \in R^{N \times N}$ in Equation 5 represents the normalization weights. Equations 6 and 5 provide recurrent feedback so that the network is almost always in a normalized state. The semi-saturation constant is set to $\sigma = 0.15$ in this work, although adaptation modulates the effective value in practice. The single-layer form of the ORGaNICs equations [15, 16] is exactly recovered by setting the last term in the Equation 4 to zero and disregarding Equations 7 and 8; The fixed point of this reduced system closely approximates Equation 2 with $n = 2$ (see [32] for the derivation).

On the other hand, dynamic variables $\boldsymbol{v}, \boldsymbol{g} \in \mathbb{R}^K$ are inhereted from the Duong et al. adaptation framework [6, 7]. The variable $\boldsymbol{v}$ tracks projections of the principal cell activity along K marginal directions and is used to estimate variance in these many directions. These "directions" are set by the columns of $\boldsymbol{W}_{yv} \in \mathbb{R}^{N \times K}$, the synaptic weight matrix that connects principal cells $\boldsymbol{y}$ to variance interneurons $\boldsymbol{v}$. The columns of $\boldsymbol{W}_{yv}$ are fixed unit vectors that form an overcomplete frame for N dimensions. Duong et al. [7] actually developed a learning rule for slow synaptic plasticity of this weight matrix that softens the requirement of over-completeness, but we restrict our analysis in this work to the fixed frame for simplicity. The variable $\boldsymbol{\mu}$ measures the time-averaged activity (i.e., low-pass filtered) of the interneuron input current. Subtracting $\boldsymbol{\mu}$ from the actual input current $\boldsymbol{c}$ and squaring gives an estimate of the marginal variance. Thus, Equation 8 allows the gains $\boldsymbol{g}$ to adapt according to estimates of marginal variance along K directions. So each interneuron has intracellular mechanisms that (1) sense average synaptic input $\boldsymbol{\mu}$ (2) compute a squared difference of this average with momentary inputs (3) adjust neural gains by comparing this squared product to a homeostatic target $\boldsymbol{\theta_t}$.

Formally, this framework transforms the covariance of responses by enforcing target marginal variances along an overcomplete set of axes set by the frame, $\boldsymbol{W}_{yv}$. These variances are enforced via gain-modulated interneuron feedback, in which gains act as Lagrange multipliers that constrain the $\frac{N(N+1)}{2}$ degrees of freedom that constitute positive semi-definite $N \times N$ matrices - the space of all covariance matrices. It is useful to think of these gains as scale-factors that geometrically deform the input covariance to a target (Figure 5).

The gains of the interneurons dynamically adjust (Equation 8) to achieve homeostatic target values for the variances of the principal cell activity. If variance is high along a column vector of $\boldsymbol{W}_{yv}$, then a scalar multiple of that column vector is subtracted from the activity. Gains reach steady state when marginal variances equal the

fixed homeostatic target vector, $\boldsymbol{\theta_t}$. A special case of this is symmetric whitening, where $\boldsymbol{\theta_t}$ is a vector of ones [6, 7]. In general, each interneuron may have its own target variance so that any arbitrary covariance transformation may be approximated by the circuit.

Two modifications to the Duong et al. [6] adaptation framework were needed to combine the two models and allow for concurrent normalization and adaptation:

1. The addition of the prefactor $\frac{\boldsymbol{a}^+}{1+\boldsymbol{a}^+}$ to the gain feedback. This term scales gain feedback so that the overall response magnitude remains normalized throughout adaptation. This intuition is formalized further in the fixed point derivation.
2. The explicit mean-centering step in the gain dynamics of Equation 8. In Duong et al.'s framework, inputs are assumed to already be mean-centered [6]. This allows variance interneurons to track actual marginal variances $\boldsymbol{v}^2 \propto (\boldsymbol{z} - \langle \boldsymbol{z} \rangle)^2$, rather than first order statistics $\boldsymbol{v}^2 \propto \boldsymbol{z}^2$. Mean-centering the inputs is not viable in our framework, since it would artificially alter the response magnitude which would interfere with normalization. Instead, we mean-center only in the gain update so that gains $\boldsymbol{g}$ correctly update according to second-order statistics. Thus, the "variance" interneurons $\boldsymbol{v}$ are not actually themselves variance estimates. Rather, they are a projection used to estimate marginal variance.

Equations 4-8 are specifically designed so that the fixed point of the recurrent dynamics is both normalized and covariance-transformed:

$$\boldsymbol{y}^2 = \frac{\boldsymbol{z'}^2}{\sigma^2 + \boldsymbol{N}\boldsymbol{z'}^2} \tag{10}$$

$$\boldsymbol{z'} = \left[\boldsymbol{I} + \boldsymbol{W}_{yv}\boldsymbol{D}(\boldsymbol{g})\boldsymbol{W}_{yv}^\top\right]^{-1}\boldsymbol{z} \tag{11}$$

This fixed point can be interpreted as a state that is both normalized and adapted. Namely, $\boldsymbol{z'}$ is the adapted input drive that is computed via a covariance transformation on the raw inputs (Equation 11). This transformed input drive $\boldsymbol{z'}$ is specifically encoded so that the covariance of the response tends toward homeostasis. The fixed point is found when this adapted state is then normalized (Equation 10). Thus, the circuit achieves a fixed point that is both normalized and properly adapted according to the two parent frameworks [6, 15, 16].

3.3 Steady State Derivation

In this section, we walk through the fixed point derivation of the dynamical system (Equations 4-8). We proceed by constraining the recurrent weight matrix to be the identity matrix, and by assuming a separation of time scales such that the gains are approximately fixed compared to the much faster dynamics of the principal cells and interneurons. Then we derive the fixed point of the neural circuit dynamics in response to an input drive $\boldsymbol{z}$, given by Equations 10 and 11.

Because of time-scale separation, we assume fixed gains $\boldsymbol{g}$. By Equation 7, the fixed point of $\boldsymbol{v}$ is especially simple: $\boldsymbol{v}_{steady} = \boldsymbol{g} \odot \boldsymbol{W}_{yv}^\top \sqrt{\boldsymbol{y}^+}$. Plugging $\boldsymbol{v}_{steady}$ into Equations 4, 5, and 6 and setting derivatives to zero gives:

$$0 = -\boldsymbol{y} + \beta \cdot \boldsymbol{z} + \left(\frac{1}{1+\boldsymbol{a}^+}\right) \boldsymbol{W}_{yy}\sqrt{\boldsymbol{y}^+} - \left(\frac{\boldsymbol{a}^+}{1+\boldsymbol{a}^+}\right) \boldsymbol{W}_{yv}\boldsymbol{D}(\boldsymbol{g})\boldsymbol{W}_{yv}^\top\sqrt{\boldsymbol{y}^+} \tag{12}$$

$$0 = -\boldsymbol{u} + \left(\frac{\sigma}{2}\right)^2 + \boldsymbol{N}(\boldsymbol{y}_1^+ \odot \boldsymbol{u}^{+^2}) \tag{13}$$

$$0 = -\boldsymbol{a} + \boldsymbol{u}^+ + \boldsymbol{a} \odot \boldsymbol{u}^+ \tag{14}$$

To approximate membrane potentials from firing rates, we work backwards from the rectification process. Assuming that membrane potentials are dominated by nonnegative values, we approximate: $\sqrt{\boldsymbol{y}^+} \approx \boldsymbol{y}$, $\boldsymbol{a}^+ \approx \boldsymbol{a}$, and $\boldsymbol{u}^+ \approx |\boldsymbol{u}|^{1/2}$. We further assume that the recurrent weight matrix $\boldsymbol{W}_{yy}$ is the identity matrix, and manipulate Equation 12 by grouping the $\boldsymbol{y}$ terms and factoring:

$$\left[\boldsymbol{I} - \left(\frac{1}{1+\boldsymbol{a}^+}\right)\boldsymbol{I} + \left(\frac{\boldsymbol{a}^+}{1+\boldsymbol{a}^+}\right)\boldsymbol{W}_{yv}\boldsymbol{D}(\boldsymbol{g})\boldsymbol{W}_{yv}^\top\right]\boldsymbol{y} = \beta \cdot \boldsymbol{z} \tag{15}$$

Simplifying and factoring again:

$$\left(\frac{\boldsymbol{a}^+}{1+\boldsymbol{a}^+}\right)\left[\boldsymbol{I} + \boldsymbol{W}_{yv}\boldsymbol{D}(\boldsymbol{g})\boldsymbol{W}_{yv}^\top\right]\boldsymbol{y} = \beta \cdot \boldsymbol{z} \tag{16}$$

In this form, it is clear why the prefactor $\left(\frac{\boldsymbol{a}^+}{1+\boldsymbol{a}^+}\right)$ is necessary. Mathematically, it scales the gain feedback so that the inverse of the covariance transformation matrix $\left[\boldsymbol{I} + \boldsymbol{W}_{yv}\boldsymbol{D}(\boldsymbol{g_{opt}})\boldsymbol{W}_{yv}^\top\right]$ can be factored out.
Making the approximation that the variable $\boldsymbol{a}$ is uniform, which is exactly true if $\boldsymbol{N}$ is the matrix of all ones, we are able to multiply by the inverse of the matrix in brackets to get:

$$\left(\frac{\boldsymbol{a}^+}{1+\boldsymbol{a}^+}\right)\boldsymbol{y} = \left[\boldsymbol{I} + \boldsymbol{W}_{yv}\boldsymbol{D}(\boldsymbol{g})\boldsymbol{W}_{yv}^\top\right]^{-1}\beta \cdot \boldsymbol{z} \tag{17}$$

Solving Eqn 14 for $\boldsymbol{u}$ in terms of $\boldsymbol{a}$ gives the very simple relation:

$$|\boldsymbol{u}|^{1/2} = \frac{\boldsymbol{a}}{1+\boldsymbol{a}} \tag{18}$$

Approximating $\boldsymbol{a}^+ \approx \boldsymbol{a}$, we can substitute this into the left side of Eqn 17 and define a modified stimulus $\boldsymbol{z'}$ as the linear gain-transformation of the raw stimulus:

$$|\boldsymbol{u}|^{1/2}\boldsymbol{y} = \beta\boldsymbol{z'} \text{ where } \boldsymbol{z'} = \left[\boldsymbol{I} + \boldsymbol{W}_{yv}\boldsymbol{D}(\boldsymbol{g})\boldsymbol{W}_{yv}^\top\right]^{-1}\boldsymbol{z} \tag{19}$$

This equation allows us to express the principal membrane potential in terms of only the modified stimulus and constants. We set $\beta = \frac{1}{2}$ and plug Eqn. 19 into Eqn. 13:

$$\frac{\sigma^2}{4} + \boldsymbol{N}\left(\frac{\boldsymbol{z'}^2}{4}\right) - \boldsymbol{z'}^2\frac{1}{4\boldsymbol{y}^2} = 0 \tag{20}$$

Finally, we perform simple algebra to solve for the steady-state responses of the principal cells. We find that it is the normalized, covariance-transformed inputs:

$$\boxed{\boldsymbol{y}^2 = \frac{\boldsymbol{z'}^2}{\sigma^2 + \boldsymbol{N}\boldsymbol{z'}^2} \; ; \; \boldsymbol{z'} = \left[\boldsymbol{I} + \boldsymbol{W}_{yv}\boldsymbol{D}(\boldsymbol{g})\boldsymbol{W}_{yv}^{\top}\right]^{-1}\boldsymbol{z}} \tag{21}$$

This equation holds for any value of gains during adaptation. Over time, gain feedback causes the marginal variances to converge to their target values and the resulting covariance across the input ensemble converges to the target covariance.

3.4 Modeling the Surround

To simulate the interplay of adaptation and normalization, we considered the difference in spatial extent between the two mechanisms in our model. Adaptation is believed to occur locally within subpopulations of neurons with overlapping receptive fields, while normalization pools activity across a much wider range of visual space [33]. To take advantage of this, we implemented a center-surround structure for our model neurons.

The classical receptive field (cRF) of a V1 neuron is defined as the spatial region where stimuli may directly drive spiking. Adjacent regions of the visual field constitute the receptive field "surround", defined as the neighboring region where stimuli can only indirectly impact a neuron's firing rate (Figure 5). Small stimuli centered on a neuron's RF drive only cRF activity, whereas large stimuli drive both the cRF and surround. Each subpopulation of neurons with overlapping cRFs has its own corresponding set of adaptive gains and variance interneurons, allowing adaptation to operate locally. Conversely, normalization operates by pooling activity over all model neurons regardless of spatial preference (Figure 5). Arranging our network in this way allowed us to selectively adapt different subsets of model neurons to compare how adaptation and normalization interact in our model. For consistency with experiment, we implement a brief delay (40 ms) of normalization signals from the surround [30]. This delay has very important implications for the dynamics during adaptation, especially for rapidly changing orientation gratings.

3.5 Covariance Target and Gain Restriction

In the parent adaptation framework of our model, the authors considered covariance whitening, the special case of adaptation where the target covariance is the identity matrix [6, 7]. In V1, however, neurons with overlapping RFs that are tuned for similar orientations will have naturally high covariance, simply because they respond to the same stimuli. Thus, whitening is not realizable.

Instead, we derive our target variances using covariance matrices that preserve the natural correlations (i.e., signal correlations) driven by corpus of input stimuli. Specifically, we derive the target covariance matrices by computing the covariance of equally likely gaussian inputs. To calculate the target marginal variance vector $\boldsymbol{\theta_t}$ given the overall target covariance $\boldsymbol{\Sigma}_{target}$, we simply project onto the interneuron subspace via frame vectors $\boldsymbol{w_i}$:

$$\theta_{t,i} = \boldsymbol{w_i}^{\top}\boldsymbol{\Sigma}_{target}\boldsymbol{w_i} \tag{22}$$

Following Duong et al. [6], we restrict gains to be non-negative:

$$\boldsymbol{g} = \max(\boldsymbol{g}, 0) \tag{23}$$

Duong et al. [6] found that this both increased stability and reduced the amplification of noise for ill-conditioned inputs. Nonnegative gains implies that adaptation is strictly suppressive (excluding facilitatory effects where adaptation in the surround can amplify responses), consistent with many experimental accounts of adaptation [17, 18, 27].

4 Discussion

Normalization and adaptation have largely been modeled in isolation, yet neither alone explains how adaptation can reshape V1 responses across stimuli of different sizes, durations, and statistics. We developed a recurrent circuit model in which the two computations are dynamically coupled: normalization interneurons regulate recurrent feedback, while gain-modulated interneurons factorize a covariance transformation that drives response statistics toward a target. Because the adaptive feedback is gated by the same modulator cells that implement normalization, the circuit's fixed point is both normalized and adapted at every stage of adaptation (Equations 10 and 11). Using a single set of generic model parameters, the model reproduced adaptation-induced changes in response gain and contrast gain, facilitation, tuning repulsion and attraction that depend on adaptor size and duration, population suppression patterns under biased stimulus ensembles, and the power law relating adaptation magnitude to stimulus probability. The model further predicts that this power law breaks down at extreme probability ratios because normalization bounds response magnitude.

4.1 Coupling adaptation to normalization

Our central contribution is a principled method for joining adaptation and normalization within a single dynamical circuit. Two structural modifications to the adaptation framework of Duong et al. [6] were sufficient. First, gating the adaptive feedback by the prefactor $\boldsymbol{u}^+$ allows the covariance transformation to factor out of the steady-state equations, so that responses remain normalized throughout adaptation. Second, mean-centering only within the gain update (Equation 8) allows the gains $\boldsymbol{g}$ to track second-order statistics without altering the response magnitudes that normalization regulates. The result is a single fixed point that satisfies both parent frameworks simultaneously, rather than an adaptation stage followed by a separate normalization stage. Our implementation of a covariance target with built-in redundancy also differs from the original implementation of symmetric whitening [6].

Several of our qualitative results do not depend on covariance homeostasis specifically. Any properly coupled model that combines local adaptive suppression with spatially broad normalization should theoretically produce contrast and response gain changes. Normalization is a well-established canonical computation [2], whereas the quantity that cortical adaptation holds homeostatic remains debated [8, 17]. We therefore view the coupling, rather than the specific adaptation rule, as the more general result. Our derivation suggests that any adaptive feedback onto principal neurons must be gated by the normalization modulator cells for the circuit to reach a fixed

point that is both normalized and adapted. Because cortical responses are shaped by a combination of feedforward and recurrent drive [34], this constraint is relevant to any future circuit model that seeks to unify the two computations.

Other adaptation rules fit naturally within the same architecture. In variance normalization, each principal neuron (or pool of similarly tuned principal neurons) would project onto a single interneuron whose gain adapts to hold that neuron's variance at a target value. This corresponds to setting $\boldsymbol{W_{yv}} = \boldsymbol{I}$, so that $K = N$, and would affect correlations only indirectly. Recurrent first-order equalization would correspond to the same connectivity as variance normalization but require interneuron gains to adjust according to mean-response targets rather than variance. Decorrelation, like the full covariance transformation, is also a linear transformation that can be factorized. Interneurons would read out the dominant axes of neural correlation and feed back to suppress correlations above a target, without necessarily equalizing variances. The first two alternatives require far fewer interneurons than the overcomplete frame used here (see Section 4.3), and all three could be tested against the experiments simulated in this work.

4.2 Experimentally testable predictions

The model makes several predictions, which we order from the most to the least directly testable. Predictions made by the two parent frameworks are outlined in their corresponding literature [6, 32, 35], so we focus instead on predictions that strictly pertain to the joint model.

First, the model predicts that the power law in Equation 1 breaks down at extreme stimulus probability ratios (Figure 4). Within the experimentally tested range, $\log_{10}[p_X(s_i)/p_Y(s_i)] \in [-1, 1]$, simulated responses closely followed the power law ($R^2 = 0.91$), but extending the range to $[-2, 1]$ reduced the fit to $R^2 = 0.77$. Because normalization bounds response magnitude, responses to rare stimuli cannot grow as steeply as the power law requires. This prediction can be tested with the protocol of Tring et al. [18] by adding stimulus ensembles with more extreme probability ratios.

Second, because the breakdown arises from normalization, the model predicts that it is more pronounced for large stimuli, which recruit strong normalization signals from the surround, than for small stimuli confined to the cRF. Comparing power-law fits across stimulus sizes would distinguish a normalization-limited account of adaptation magnitude from one that depends on stimulus probability alone.

Third, adaptive gains respond to mean-subtracted, second-order statistics (Equation 8). The model therefore attributes the suppression of neurons tuned orthogonal to the adaptor (Figure 3c) to the high variance of their intermittent responses rather than to their mean activity. A direct test would compare adaptation to two ensembles that evoke similar mean responses but different response variances, for example a stimulus held at moderate contrast versus the same stimulus alternating between high and zero contrast. Our model predicts stronger suppression for the high-variance ensemble, whereas models that hold mean responses homeostatic predict similar suppression for both.

Fourth, the model posits a dedicated population of interneurons whose gains track response variance along axes in population space and whose feedback is gated by

the normalization modulator cells. Selectively silencing this population should abolish adaptation-induced affects while leaving normalization, including contrast saturation and surround suppression, intact. Testing this prediction requires first identifying candidate cell types, which the model does not yet specify (Section 4.3).

4.3 Limitations as a biophysical model

The most significant limitation is the number of interneurons required. To represent an arbitrary covariance transformation, the overcomplete frame $\boldsymbol{W_{yv}}$ requires $K \geq N(N+1)/2$ interneurons, which scales as $O(N^2)$. Cortical circuits do not contain interneurons on this scale, but three simple modifications could substantially reduce the requirement. First, statistical dependencies in visual inputs decay with distance, so adaptation can be restricted to local pools of neurons with overlapping or adjacent receptive fields [6, 35, 36]. We adopted a version of this approach by assigning each subpopulation with overlapping cRFs its own set of variance interneurons (Section 3.4), which drastically reduces the total number of interneurons; within each pool, however, the frame remains overcomplete. Second, Duong et al. [7] showed that slow synaptic plasticity of $\boldsymbol{W_{yv}}$, driven by a local learning rule, achieves the same adaptation with as little as $K = N$ interneurons. Third, adaptation need not resolve every neuron individually. Neurons with overlapping receptive fields and similar orientation preferences respond to nearly the same stimuli and therefore covary strongly (Section 3.5). Variance interneurons could instead pool activity across groups of similarly tuned neurons and feed back uniformly onto each group. The covariance transformation would then act on M group-averaged responses rather than on N individual neurons, requiring only $K \geq M(M+1)/2$ interneurons with $M \ll N$. The cost of this coarse graining is that activity patterns within a group would not trigger adaptation.

Second, the model violates Dale's law, which requires each neuron to be exclusively excitatory or inhibitory at all of its synapses. The columns of $\boldsymbol{W_{yv}}$ are not sign-constrained, so each variance interneuron computes a weighted sum of principal-neuron responses with both positive and negative weights. Following Pal et al. [32], the weight matrices can be interpreted as effective connectivity, with negative weights implemented by disynaptic inhibition through interneurons that are not modeled explicitly, at the cost of additional delays. Alternatively, the frame can be constrained to be non-negative, although the resulting responses would generally be less well aligned with the target distribution [35]. The related circuit model of Lipshutz and Simoncelli [35] and hierarchical ORGaNICs [32] share this limitation. Lipshutz and Simoncelli also noted that a more realistic circuit would include nonlinear principal neurons; our model addresses this by incorporating normalization, but it does not resolve the sign constraint.

The normalization subcircuit inherits the biological interpretation of ORGaNICs [16, 32]. Pal et al. [32] hypothesize that principal neurons correspond to pyramidal neurons and that the inhibitory modulator cells $\boldsymbol{a}$ correspond to a combination of parvalbumin-expressing (PV+) and somatostatin-expressing (SST+) interneurons, consistent with the broad spatial and orientation pooling of the normalization signal. In this circuit, normalization reduces recurrent amplification as input strength increases, decreasing both recurrent excitatory and inhibitory signals, consistent with

experimental measurements of synaptic conductances [20]. The excitatory modulator cells $\boldsymbol{u}$ mediate the drive from principal neurons to $\boldsymbol{a}$, whereas pyramidal neurons excite PV+ and SST+ cells directly; however, the equations can be rewritten without this intermediate population without changing the circuit's function [32].

The adaptation subcircuit is less constrained by biology. We have not proposed a mapping of the variance interneurons $\boldsymbol{v}$ onto a specific cell type, and the model requires several multiplicative operations whose cellular mechanisms remain speculative: the gain modulation $\boldsymbol{g} \odot \boldsymbol{W}_{\boldsymbol{yv}}^{\top} \sqrt{\boldsymbol{y}^{+}}$ (Equation 7), the gating of adaptive feedback by $\boldsymbol{a}^{+}/(1+\boldsymbol{a}^{+})$ (Equation 4), and the squared deviation from a running mean that estimates marginal variance (Equation 8). For analogous element-wise products in ORGaNICs, Pal et al. [32] propose coincident spike detection or NMDA receptors, whose synaptic current depends approximately multiplicatively on presynaptic firing rate and postsynaptic depolarization. Several mechanisms of cortical gain modulation have also been identified [13], any of which could in principle implement the interneuron gains. Finally, the time constants of the principal neurons and modulator cells (0.1–0.2 ms) are unrealistically short for individual neurons. As for hierarchical ORGaNICs [32], they are better interpreted as effective population time constants. Because Equations 10 and 11 hold for any fixed gains, lengthening the fast time constants would leave the fixed point unchanged, provided they remain much shorter than , although it would alter the transient dynamics relative to the 40 ms surround delay.

Our implementations of normalization and spatial structure were deliberately simplified. Normalization pools were uniform in both spatial position and orientation preference, receptive fields of different subpopulations did not overlap, and adaptation acted identically on every receptive field. In cortex, normalization pools are unlikely to be uniform [37, 38], receptive fields overlap [39, 40], and adaptation need not affect all receptive fields equally [41]. One consequence is that surround suppression in our model was strong enough that we modeled ensemble adaptation experiments with small stimuli rather than the larger stimuli used experimentally. We did not seek to fit data. Instead, we used the same generic parameters for every simulation to show that the qualitative behavior arises from the coupling of adaptation and normalization rather than from parameter tuning. Quantitative comparison with data would require fitting the model, including the spatial and orientation profiles of the normalization pool.

Finally, the model adapts only to second-order statistics. The visual system is sensitive to higher-order statistics and extracts features more complex than the mean and covariance of its inputs [42, 43], so covariance homeostasis is necessarily an incomplete account of adaptation. Linear transformations can remove only second-order statistical dependencies [35], and the adaptation subcircuit implements exactly such a transformation (Equation 11). Normalization, however, supplies a nonlinearity: our circuit applies an adaptive linear transformation followed by divisive normalization (Equations 10 and 11). Divisive normalization can itself reduce higher-order dependencies in natural signals [44], and linear–nonlinear stages stacked into layers can, in principle, extract and modulate higher-order features. A hierarchical form of ORGaNICs already exists [32], and its authors identify adaptation as a missing component, so embedding our mechanism in a multi-layer circuit is a natural extension. Alternatively,

the nonlinearity can be placed within the adaptation subcircuit itself: Lipshutz and Simoncelli [35] showed that adapting interneuron activation functions, together with Hebbian plasticity of the connecting weights, reshapes response distributions beyond second order.

4.4 Relationship to previous models

Among recurrent circuit models of adaptation [9, 10], ours is distinguished by its coupling to normalization. Adaptation is performed by dedicated interneurons that measure population activity along many axes and feed back onto principal neurons, following Duong et al. [6]. What is new is that this adaptive feedback is gated by normalization, which is itself implemented recurrently. This coupling produces several of our results, including surround facilitation, tuning attraction after prolonged adaptation to large stimuli, and the breakdown of the power law, none of which an independently acting adaptation mechanism would capture.

The main previous model to combine adaptation with normalization is that of Westrick et al. [8], which adjusts the normalization weights between pairs of neurons based on whether the product of their responses exceeded a target. Some recurrent models similarly explain adaptation-induced tuning shifts through changes in recurrent excitatory and inhibitory weights [10]. Duong et al. [3] showed that modulation of input gains combined with recurrent feedback is sufficient to demonstrate tuning changes in a single-layer linear model. In contrast, our model adapts by modulating interneuron gains while synaptic weights and input gain remain fixed. Gain modulation in our model is fast and reversible [13], consistent with the timescale of rapid cortical adaptation [12], and avoids the instability that continuous weight remapping can introduce.

Our model also accounts for how adaptation modulates contrast-dependent response properties. Westrick et al. [8] did not model adaptation effects that arise from varying stimulus contrast, and suggested that a separate, complementary process adjusting the semi-saturation constant σ could be added to capture changes in contrast gain. In our model, these changes emerge from the same mechanism that produces pattern adaptation. Adaptation to small stimuli reduced response gain, whereas adaptation to large stimuli shifted the contrast response function rightward and reduced contrast gain (Figure 1b), consistent with experimental results [23–25]. Within the normalization model, this shift corresponds to an increase in the effective semi-saturation constant (Equation 2). In our circuit, however, σ is fixed; the increase arises from properly coupled adaptation and normalization mechanisms rather than from a separately adapted parameter. Adaptation of the surround alone produced a separate effect, facilitating responses above the unadapted baseline [25]. The model thus qualitatively accounts for contrast and pattern adaptation with a single mechanism, consistent with evidence that the two share a common gain control mechanism [45].

An efficient-coding account [4] proposed that tuning repulsion and attraction follow from a single coding principle under different prior conditions. For example, populations optimized for a wide stimulus prior exhibit tuning attraction toward overrepresented stimuli, whereas sharp priors typically produce repulsion [4]. Our results provide a complementary mechanistic account: whether fast adaptation produces

repulsion or attraction also depends strongly on its interaction with normalization, in particular orientation-specific surround suppression. Adaptation within the cRF produced repulsion, whereas adaptation of the surround released neurons from normalization for certain stimuli and produced attraction (Figures 1c and 2). This account is consistent with experimental evidence that, depending on the size of the stimulus, adaptation can cause tuning repulsion or attraction [27] [46] .

Our model inherits a mechanistic account of two efficient-coding principles: variance normalization and redundancy reduction. Previous descriptions of efficient adaptation, including the power law in Equation 1, were derived normatively or characterized empirically without specifying a circuit implementation [5, 18]. In our model, these principles are enforced without a global learning rule, and the power law emerged from local gain dynamics without being designed into the model.

Finally, our results may be relevant to machine learning. Biological visual systems remain far more energy-efficient and data-efficient than state-of-the-art artificial systems [47, 48], and biologically inspired adaptation and normalization have improved robustness and generalization of artificial networks in specific settings [49]. Decorrelation and whitening are common preprocessing steps in machine learning [50]. Our circuit performs them dynamically and online while normalization keeps responses within a stable range. ORGaNICs were proposed as a biophysically plausible counterpart to long short-term memory networks [15], and their recurrent normalization confers stability even with strong recurrent connections [51, 52]. Whether jointly adaptive and normalized units improve the efficiency or robustness of artificial networks remains to be tested.

In summary, we have shown that adaptation and normalization can be implemented jointly in a single recurrent circuit whose fixed point preserves the core function of each. To the extent that normalization is a canonical neural computation [2], the same coupling may govern adaptation along other stimulus dimensions and in other sensory areas.

References

[1] Heeger, D.J.: Normalization of cell responses in cat striate cortex. Visual Neuroscience **9**(2), 181–197 (1992) https://doi.org/10.1017/S0952523800009640 . Accessed 2026-10-27

[2] Carandini, M., Heeger, D.J.: Normalization as a canonical neural computation. Nature Reviews Neuroscience **13**(1), 51–62 (2012) https://doi.org/10.1038/nrn3136 . Accessed 2026-07-21

[3] Duong, L.R., Bredenberg, C., Heeger, D.J., Simoncelli, E.P.: Adaptive coding efficiency in recurrent cortical circuits via gain control. arXiv. arXiv:2305.19869 [q-bio] (2023). https://doi.org/10.48550/arXiv.2305.19869 . http://arxiv.org/abs/2305.19869 Accessed 2026-01-26

[4] Prat-Carrabin, A., Harl, M.V., Gershman, S.J.: Fast efficient coding and sensory adaptation in gain-adaptive recurrent networks. Nature Communications **17**(1),

5554 (2026) https://doi.org/10.1038/s41467-026-73032-0 . Accessed 2026-06-29

[5] Mariani, M., Moosavi, A.S., Ringach, D., Dipoppa, M.: A universal power law optimizes energy and representation fidelity in visual adaptation. Neuroscience (2025). https://doi.org/10.1101/2025.03.20.643406 . http://biorxiv.org/lookup/doi/10.1101/2025.03.20.643406 Accessed 2026-07-13

[6] Duong, L.R., Lipshutz, D., Heeger, D.J., Chklovskii, D.B., Simoncelli, E.P.: Adaptive whitening in neural populations with gain-modulating interneurons. arXiv. arXiv:2301.11955 [q-bio] (2023). https://doi.org/10.48550/arXiv.2301.11955 . http://arxiv.org/abs/2301.11955 Accessed 2026-01-26

[7] Duong, L.R., Simoncelli, E.P., Chklovskii, D.B., Lipshutz, D.: Adaptive whitening with fast gain modulation and slow synaptic plasticity. arXiv. arXiv:2308.13633 [q-bio] (2023). https://doi.org/10.48550/arXiv.2308.13633 . http://arxiv.org/abs/2308.13633 Accessed 2026-01-26

[8] Westrick, Z.M., Heeger, D.J., Landy, M.S.: Pattern Adaptation and Normalization Reweighting. The Journal of Neuroscience **36**(38), 9805–9816 (2016) https://doi.org/10.1523/JNEUROSCI.1067-16.2016 . Accessed 2026-05-15

[9] Mohan, V., Rideaux, R.: Energy efficiency and sensitivity benefits in a motion processing adaptive recurrent neural network (2024). https://doi.org/10.1101/2024.12.20.629845 . http://biorxiv.org/lookup/doi/10.1101/2024.12.20.629845 Accessed 2026-02-09

[10] Teich, A.F., Qian, N.: Learning and Adaptation in a Recurrent Model of V1 Orientation Selectivity. Journal of Neurophysiology **89**(4), 2086–2100 (2003) https://doi.org/10.1152/jn.00970.2002 . Accessed 2026-07-08

[11] Abbott, L.F., Varela, J.A., Sen, K., Nelson, S.B.: Synaptic Depression and Cortical Gain Control. Science **275**(5297), 221–224 (1997) https://doi.org/10.1126/science.275.5297.221 . Accessed 2026-01-26

[12] Müller, J.R., Metha, A.B., Krauskopf, J., Lennie, P.: Rapid Adaptation in Visual Cortex to the Structure of Images. Science **285**(5432), 1405–1408 (1999) https://doi.org/10.1126/science.285.5432.1405 . Accessed 2026-10-27

[13] Ferguson, K.A., Cardin, J.A.: Mechanisms underlying gain modulation in the cortex. Nature Reviews Neuroscience **21**(2), 80–92 (2020) https://doi.org/10.1038/s41583-019-0253-y . Accessed 2026-10-27

[14] Heeger, D.J., Mackey, W.E.: ORGaNICs: A Theory of Working Memory in Brains and Machines. arXiv. arXiv:1803.06288 [cs] (2018). https://doi.org/10.48550/arXiv.1803.06288 . http://arxiv.org/abs/1803.06288 Accessed 2026-01-26

[15] Heeger, D.J., Mackey, W.E.: Oscillatory recurrent gated neural integrator circuits

(ORGaNICs), a unifying theoretical framework for neural dynamics. Proceedings of the National Academy of Sciences **116**(45), 22783–22794 (2019) https://doi.org/10.1073/pnas.1911633116 . Accessed 2026-01-26

[16] Heeger, D.J., Zemlianova, K.O.: A recurrent circuit implements normalization, simulating the dynamics of V1 activity. Proceedings of the National Academy of Sciences **117**(36), 22494–22505 (2020) https://doi.org/10.1073/pnas.2005417117 . Accessed 2026-01-26

[17] Benucci, A., Saleem, A.B., Carandini, M.: Adaptation maintains population homeostasis in primary visual cortex. Nature Neuroscience **16**(6), 724–729 (2013) https://doi.org/10.1038/nn.3382 . Accessed 2026-02-12

[18] Tring, E., Dipoppa, M., Ringach, D.L.: A power law describes the magnitude of adaptation in neural populations of primary visual cortex. Nature Communications **14**(1), 8366 (2023) https://doi.org/10.1038/s41467-023-43572-w . Accessed 2026-06-24

[19] Adesnik, H.: Synaptic Mechanisms of Feature Coding in the Visual Cortex of Awake Mice. Neuron **95**(5), 1147–11594 (2017) https://doi.org/10.1016/j.neuron.2017.08.014 . Accessed 2026-09-23

[20] Ozeki, H., Finn, I.M., Schaffer, E.S., Miller, K.D., Ferster, D.: Inhibitory Stabilization of the Cortical Network Underlies Visual Surround Suppression. Neuron **62**(4), 578–592 (2009) https://doi.org/10.1016/j.neuron.2009.03.028 . Accessed 2026-09-23

[21] Solomon, S., Kohn, A.: Moving Sensory Adaptation beyond Suppressive Effects in Single Neurons. Current Biology **24**(20), 1012–1022 (2014) https://doi.org/10.1016/j.cub.2014.09.001 . Accessed 2026-05-15

[22] Kohn, A.: Visual Adaptation: Physiology, Mechanisms, and Functional Benefits. Journal of Neurophysiology **97**(5), 3155–3164 (2007) https://doi.org/10.1152/jn.00086.2007 . Accessed 2026-05-15

[23] Ohzawa, I., Sclar, G., Freeman, R.D.: Contrast gain control in the cat's visual system. Journal of Neurophysiology **54**(3), 651–667 (1985) https://doi.org/10.1152/jn.1985.54.3.651 . Accessed 2026-07-21

[24] Crowder, N.A., Price, N.S.C., Hietanen, M.A., Dreher, B., Clifford, C.W.G., Ibbotson, M.R.: Relationship Between Contrast Adaptation and Orientation Tuning in V1 and V2 of Cat Visual Cortex. Journal of Neurophysiology **95**(1), 271–283 (2006) https://doi.org/10.1152/jn.00871.2005 . Accessed 2026-07-21

[25] Dhruv, N.T., Tailby, C., Sokol, S.H., Lennie, P.: Multiple adaptable mechanisms early in the primate visual pathway. The Journal of Neuroscience: The Official Journal of the Society for Neuroscience **31**(42), 15016–15025 (2011) https://doi.

org/10.1523/JNEUROSCI.0890-11.2011

[26] Patterson, C.A., Duijnhouwer, J., Wissig, S.C., Krekelberg, B., Kohn, A.: Similar adaptation effects in primary visual cortex and area MT of the macaque monkey under matched stimulus conditions. Journal of Neurophysiology **111**(6), 1203–1213 (2014) https://doi.org/10.1152/jn.00030.2013

[27] Patterson, C.A., Wissig, S.C., Kohn, A.: Distinct effects of brief and prolonged adaptation on orientation tuning in primary visual cortex. The Journal of Neuroscience: The Official Journal of the Society for Neuroscience **33**(2), 532–543 (2013) https://doi.org/10.1523/JNEUROSCI.3345-12.2013

[28] Henry, C.A., Joshi, S., Xing, D., Shapley, R.M., Hawken, M.J.: Functional Characterization of the Extraclassical Receptive Field in Macaque V1: Contrast, Orientation, and Temporal Dynamics. The Journal of Neuroscience **33**(14), 6230–6242 (2013) https://doi.org/10.1523/JNEUROSCI.4155-12.2013 . Accessed 2026-10-09

[29] Smith, M.A., Bair, W., Movshon, J.A.: Dynamics of Suppression in Macaque Primary Visual Cortex. The Journal of Neuroscience **26**(18), 4826–4834 (2006) https://doi.org/10.1523/JNEUROSCI.5542-06.2006 . Accessed 2026-10-09

[30] Bair, W., Cavanaugh, J.R., Movshon, J.A.: Time Course and Time-Distance Relationships for Surround Suppression in Macaque V1 Neurons. The Journal of Neuroscience **23**(20), 7690–7701 (2003) https://doi.org/10.1523/JNEUROSCI.23-20-07690.2003 . Accessed 2026-10-05

[31] Carandini, M., Heeger, D.J., Movshon, J.A.: Linearity and Normalization in Simple Cells of the Macaque Primary Visual Cortex. The Journal of Neuroscience **17**(21), 8621–8644 (1997) https://doi.org/10.1523/JNEUROSCI.17-21-08621.1997 . Accessed 2026-09-23

[32] Pal, A., Rawat, S., Heeger, D.J., Martiniani, S.: Hierarchical Neural Circuit Theory of Normalization and Inter-areal Communication (2025). https://doi.org/10.1101/2025.07.15.664935 . http://biorxiv.org/lookup/doi/10.1101/2025.07.15.664935 Accessed 2026-01-26

[33] Kohn, A., Movshon, J.A.: Neuronal Adaptation to Visual Motion in Area MT of the Macaque. Neuron **39**(4), 681–691 (2003) https://doi.org/10.1016/S0896-6273(03)00438-0 . Accessed 2026-10-05

[34] Douglas, R.J., Koch, C., Mahowald, M., Martin, K.A.C., Suarez, H.H.: Recurrent Excitation in Neocortical Circuits. Science **269**(5226), 981–985 (1995) https://doi.org/10.1126/science.7638624 . Accessed 2026-09-23

[35] Lipshutz, D., Simoncelli, E.P.: Shaping the distribution of neural responses with interneurons in a recurrent circuit model (2024) https://doi.org/10.48550/

ARXIV.2405.17745 . Version Number: 2. Accessed 2026-11-04

[36] Wainwright, M.J., Shwartz, O., Simoncelli, E.P.: Natural image statistics and divisive normalization: Modeling nonlinearity and adaptation in cortical neurons. In: Rao, R.P.N., Olshausen, B.A., Lewicki, M.S. (eds.) Probabilistic Models of the Brain: Perception and Neural Function / Edited by Rajesh P.N. Rao, Bruno A. Olshausen, Michael S. Lewicki. Neural information processing series. MIT Press, Cambridge, Mass (2002)

[37] Burg, M.F., Cadena, S.A., Denfield, G.H., Walker, E.Y., Tolias, A.S., Bethge, M., Ecker, A.S.: Learning divisive normalization in primary visual cortex. PLOS Computational Biology **17**(6), 1009028 (2021) https://doi.org/10.1371/journal.pcbi.1009028 . Accessed 2026-11-05

[38] Cavanaugh, J.R., Bair, W., Movshon, J.A.: Selectivity and Spatial Distribution of Signals From the Receptive Field Surround in Macaque V1 Neurons. Journal of Neurophysiology **88**(5), 2547–2556 (2002) https://doi.org/10.1152/jn.00693.2001 . Accessed 2026-11-05

[39] Hubel, D.H., Wiesel, T.N.: Uniformity of monkey striate cortex: A parallel relationship between field size, scatter, and magnification factor. Journal of Comparative Neurology **158**(3), 295–305 (1974) https://doi.org/10.1002/cne.901580305 . Accessed 2026-11-05

[40] Smith, S.L., Häusser, M.: Parallel processing of visual space by neighboring neurons in mouse visual cortex. Nature Neuroscience **13**(9), 1144–1149 (2010) https://doi.org/10.1038/nn.2620 . Accessed 2026-11-05

[41] Marlin, S.G., Douglas, R.M., Cynader, M.S.: Position-specific adaptation in simple cell receptive fields of the cat striate cortex. Journal of Neurophysiology **66**(5), 1769–1784 (1991) https://doi.org/10.1152/jn.1991.66.5.1769 . Accessed 2026-11-05

[42] Freeman, J., Ziemba, C.M., Heeger, D.J., Simoncelli, E.P., Movshon, J.A.: A functional and perceptual signature of the second visual area in primates. Nature Neuroscience **16**(7), 974–981 (2013) https://doi.org/10.1038/nn.3402 . Accessed 2026-11-05

[43] Purpura, K.P., Victor, J.D., Katz, E.: Striate cortex extracts higher-order spatial correlations from visual textures. Proceedings of the National Academy of Sciences **91**(18), 8482–8486 (1994) https://doi.org/10.1073/pnas.91.18.8482 . Accessed 2026-11-05

[44] Schwartz, O., Simoncelli, E.P.: Natural signal statistics and sensory gain control. Nature Neuroscience **4**(8), 819–825 (2001) https://doi.org/10.1038/90526 . Accessed 2026-11-05

[45] Moosavi, S.A., Tring, E., Ringach, D.L.: Contrast and pattern adaptation in visual cortex share a common gain control mechanism. Journal of Neurophysiology **135**(6), 1467–1479 (2026) https://doi.org/10.1152/jn.00567.2025 . Accessed 2026-06-24

[46] Aschner, A., Solomon, S.G., Landy, M.S., Heeger, D.J., Kohn, A.: Temporal Contingencies Determine Whether Adaptation Strengthens or Weakens Normalization. The Journal of Neuroscience **38**(47), 10129–10142 (2018) https://doi.org/10.1523/JNEUROSCI.1131-18.2018 . Accessed 2026-01-26

[47] Mehonic, A., Kenyon, A.J.: Brain-inspired computing needs a master plan. Nature **604**(7905), 255–260 (2022) https://doi.org/10.1038/s41586-021-04362-w . Accessed 2026-11-05

[48] Huber, L.S., Geirhos, R., Wichmann, F.A.: The developmental trajectory of object recognition robustness: Children are like small adults but unlike big deep neural networks. Journal of Vision **23**(7), 4 (2023) https://doi.org/10.1167/jov.23.7.4 . Accessed 2026-11-05

[49] Vinken, K., Boix, X., Kreiman, G.: Incorporating intrinsic suppression in deep neural networks captures dynamics of adaptation in neurophysiology and perception. Science Advances **6**(42), 4205 (2020) https://doi.org/10.1126/sciadv.abd4205 . Accessed 2026-11-05

[50] Huang, L., Yang, D., Lang, B., Deng, J.: Decorrelated Batch Normalization. In: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 791–800. IEEE, Salt Lake City, UT, USA (2018). https://doi.org/10.1109/CVPR.2018.00089 . https://ieeexplore.ieee.org/document/8578187/ Accessed 2026-11-05

[51] Rawat, S., Heeger, D., Martiniani, S.: Unconditional stability of a recurrent neural circuit implementing divisive normalization. In: Advances in Neural Information Processing Systems 37, pp. 14712–14750. Neural Information Processing Systems Foundation, Inc. (NeurIPS), Vancouver, BC, Canada (2024). https://doi.org/10.52202/079017-0470 . http://www.proceedings.com/079017-0470.html Accessed 2026-11-05

[52] Morone, F., Rawat, S., Heeger, D.J., Martiniani, S.: Stabilization of recurrent neural networks through divisive normalization. Proceedings of the National Academy of Sciences **123**(30), 2601841123 (2026) https://doi.org/10.1073/pnas.2601841123 . Accessed 2026-11-05